



# EMP 211

.



# CURRICULUM CONTEXT



## Definitions of basic concepts in curriculum

### Education

The term education mainly refers to knowledge, skills and attitudes one acquires either through formal schooling or informal means. Oluoch, (1982) defines **Education** as the process of acquiring and developing desired knowledge, skills and attitudes. Other terms associated with education include **Instruction**, schooling, and learning.

What is the difference between?

- Education and instruction
- Education and schooling
- Education and learning

The main purpose of education is to socialize an individual among peer groups. From the beginning of human society, the main objective of education has been that of transmitting to the child the accumulated experiences of his people and their culture as well as of training him to fit into the membership of the group. It was through this background that the society has managed to survive and gain increasing knowledge over all forms of human institutions: governance, rituals, survival etc.

### Curriculum

There are many meanings attached to the word —curriculum sometimes it is referred to as —syllabus or list of subjects or course of study or topics or terms of knowledge to be covered or content or organization of teaching and learning or method or time table etc. The term has therefore been defined differently by various authors. Amongst the most notable include the following:

- Tanner (1980) defined curriculum as —the planned and guided learning experiences and intended outcomes, formulated through the systematic reconstruction of knowledge and experiences under the auspices of the school, for the learners' continuous and willful growth in personal social competence.
- Pratt (1980) defines curriculum as a written document that systematically describes goals planned, objectives, content, learning activities, evaluation procedures and so forth.
- Goodlad and Su (1992) define curriculum as a plan that consists of learning opportunities for a specific time frame and place, a tool that aims to bring about behaviour changes in students as a result of planned activities and includes all learning experiences received by students with the guidance of the school.
- Hass (1987) provides a broader definition, stating that a curriculum includes —all of the experiences that individual learners have in a program of education whose purpose is to achieve broad goals and related specific objectives, which is planned in terms of a framework of theory and research or past and present professional practice (p.5).
- Oluoch (1982) defines curriculum as all that is planned to enable the students acquire and develop the desired knowledge, skills and attitudes

Curriculum is a plan of Education. This plan transforms ideas into curriculum which is related to life, needs, aspirations and problems of people. In this manner, the curriculum becomes a powerful and dynamic instrument of social, economic and cultural transformation of the society

Curriculum is much wider than syllabus, where a syllabus is only part of the total curriculum. So, curriculum is concerned not so much with prescribing the knowledge to be acquired as with the area of learning experiences to be organized by teachers, both within and outside the school to enable pupils to adopt a positive attitude to learning, but also to acquire and apply knowledge and skills to develop pupils' tastes and a balanced sense of values.

Curriculum was prompted during the World Wars; especially the 1st World War (1918) when Franklin Bobbitt published a book called —the CurriculumI to provide a course of operations on the war and it's derived out comes.

This period was marked by industrial and technological development. There was need to design programmes of activity or events which had profound effect on the social, economic and political life of people. Curriculum as seen now is to create a situation of social activity- based on development of the people.

### **Competency Based Curriculum**

What is competency based curriculum?

- Competency can be defined as the ability to do a particular activity to a prescribed standard
- In CBC, the emphasis is on what students need to know and be able to do with emphasis being on what they can do in varying and complex real life situations (problem solving)
- Therefore CBE is outcome / learning-based rather than content-based education

- In CBE, the goal is to get ALL students to master a set of skills to a predetermined standard – research has shown that this is possible given sufficient time and resources. And hence one of the intentions of the CBC is stated as: ensuring all learners are successful and become engaged, empowered and ethical citizens through its mission of nurturing every learner's potential.

| FROM                                                       | TO                                                                |
|------------------------------------------------------------|-------------------------------------------------------------------|
| Content Focus                                              | Focus on competencies                                             |
| Rigid and prescriptive curriculum with limited flexibility | Flexible with opportunities for specialization                    |
| Focus on summative assessment and competition              | Balance between formative and summative assessment and excellence |
| Emphasis on schooling                                      | Emphasis on education                                             |

## **.2.2 Basic Curriculum Questions and elements of curriculum**

Although Franklin Bobbitt (1918) laid the foundation and justification for curriculum development, the process of designing a curriculum remained a challenge to curriculum developers. Ralph Tyler (1949) in his monograph provided a rationale for developing a curriculum and has since been the main guide in curriculum development in the world. Referred to as the objective model Tyler identified four basic questions that guide the [curriculum development process](#).

- What educational purposes should the school seek to attain?
- What educational experiences should be provided to attain these purposes?
- How can these experiences be effectively organized?
- How can we determine whether the purposes are being attained?

**Answers to these four basic questions therefore guide in:**

- Formulation of aims, goals and objectives of the education for a particular group/level
- Selection of experiences/content/learning activities for inclusion in the school program
- Organization, sequencing or ordering of the content/experiences/learning activities
- Evaluation of the programme

### **Elements of a school curriculum**

The four questions supply the basis of the objective model of curriculum development. These form the basic elements of curriculum, namely;

- Aims and objectives/learning outcomes
- Content/subject matter
- Organization of learning experiences/methods

- Evaluation

These four basic elements of curriculum do not constitute neat, discrete categories. They form a dynamic, organic whole and are closely interrelated and each element is influenced by, and influences, the others. However most curriculum designs contain three major elements, namely;

- **Objectives** of the curriculum/learning outcomes
- The **learning experiences** as a means of achieving those objectives

c. The **evaluation** process for appraising the suitability of the programme/measuring the extent to which objectives have been achieved. Olouch (1982) refers to this as student assessment.

### 1.2.3 Dimensions of the School Curriculum

Oluoch (1982) in his conceptual framework of curriculum development, he identifies three curriculum elements and three main dimensions of a school curriculum. Elements here refer to parts of a school curriculum while dimensions refer to its aspects. The dimensions are

#### **Formal dimension**

This refers that aspect of the school curriculum which consists of those learning activities that students undertake formally as a class as well as curriculum objectives and student assessment methods that relate to them. They are carried out as formal class work though they may be carried out within or outside the school.

#### **1. Non formal dimension**

This refers to those learning activities traditionally referred to as extra-curricular or more recently core-curricular. They are organized in less rigid manner than formal learning activities. They are not carried out by students in their regular groupings rather students group themselves in accordance with such factors as individual interests, attitudes and ages. They may be carried out inside or outside the classroom or within or without the school compound. The non-formal dimension refers to these non formal learning activities as well as the objectives and student assessment methods that relate to them

### **1. In-formal dimension**

This consists of the guided aspects of the informal learning activities that go on in the school all the time. These include the interaction with the planned aspects of the school environment, e.g., assimilation of desired habits by students from good examples deliberately given by the school.

Informal interaction goes on all the time and not all interaction forms desirable informal learning activities and only the activities that relate to the planned aspect of the environment count as nonformal dimension. The planning of the environment is done through formulation of school rules and regulations, encouragement of desirable lifestyles among the staff of an institution. The richer the environment the more the opportunities of benefitting from the curriculum dimension.

## **1.2.5 Curriculum documents**

For facilitate effective curriculum development a teacher will require to have the following documents

1. Syllabuses/curriculum designs



The syllabus/curriculum designs show the topics, learning outcomes/objectives, content and activities for each subject on the curriculum. They guide teachers in the preparation of the schemes of work

#### 1. Schemes of work

- This is an outline showing the topics to be covered in each subject in the curriculum in a given term/year
- A detailed scheme of work can
- Identify what has been covered during the term and what had been left out. A teacher can then make realistic changes for the following year
- Help teachers teaching multiple streams ensure that each class covers the same amount of work by keeping track of topics taught in each class
- Give teachers who take class during the next year a clearer idea of what areas have been covered

#### 1. Lesson plans

This is a plan or guide that a teacher uses to effectively deliver a lesson. Lesson plans can be done on a daily or weekly basis as per policy requirements. A lesson mainly contains instructional objectives/learning outcomes, the content to be covered, the resources and teaching aids that the

teacher would need to use during the lesson, the teaching method that the lesson will use to deliver the lesson, the learning activities the learners will do to and the monitoring and evaluation.

1. Textbooks/teachers guides/reference materials

Teachers will need subject textbooks for lesson preparation. Reference books and other supplementary materials are necessary for proper lesson planning.

1. Pupils assessment/progress records

This is a document with record of learner's assessment. This record is important in monitoring learner's progress

1. Class attendance register

This is a document that shows the registered learners in a class and their daily class attendance. Its used both for administrative purposes and curriculum supervision purposes

1. Record of work covered

This is a document that shows what has been covered and what has not been covered with the necessary details and comments.

# FOUNDATIONS OF CURRICULUM



## Introduction

The foundations of Curriculum comprise of values, traditions, factors and forces which influence the kind, quantity and quality of the experience the school offers to its learners. In this lesson we look at historical, philosophical, psychological, sociological and technological foundations of curriculum and their influence and implications to curriculum development and innovation.



## Lesson Learning Outcomes

By the end of this lesson, you will be able to:

- Discuss historical foundations and its influence on curriculum development
- Discuss philosophical foundations and its implications for curriculum development
- Discuss psychological foundations and its implications for curriculum development
- Discuss sociological foundations and its implications for curriculum development
- Discuss professional foundations and its implications on curriculum devbelopment Discuss technological foundations and its implications for curriculum development



## Historical Foundations

Historical foundations deals with past events, which have led to present trends in education to

inform curriculum decisions in terms of current and future education needs.

The study of history of education can help us understand why certain practices have evolved over time. It can help us understand the controversies that have faced educational leaders in the past and the choices they made. This understanding can help us understand current controversies, which can help us choose among competing proposals. The study of history can help us uncover practices that have been successful and those that have been not.

History of education provides a context and case studies that can help us evaluate current educational practices and proposals for change. In reviewing history we need to consider:

- Why a particular educational practice was established?
- Does that practice still meet a need?
- Does the proposal for change meet new needs or is it likely to take us in directions that are inconsistent with our deeply held values?

A basic purpose for the study of history of education is that it provides a rich understanding of the issues we need to consider as we make decisions about change and the future. Curriculum developers always ensure the historical perspective is well reflected when designing curriculum in order to capture not only the local flavour but also global historical views.

### **Historical developments that have shaped Kenya education system**

- The European Influence

Many of the school practices are so familiar and reflect long standing cultural choices. Consider the following characteristics of our education (Armstrong, Henson and Savage, 2009)

- The idea that concepts should be organised under major categories or headings
- The idea that knowledge should be divided among individual subjects
- The idea that teaching should occur in a setting that brings young people together in groups for instructional purposes
- The idea that schools should be organised in sequence of grades
- The idea that as many people as possible should be educated
- The idea that teachers should consider individual differences when planning instruction
- The idea that schools should help learners develop rational thinking processes
- The idea that teachers should have some kind of specialised training
- The idea that schooling should be a preparation for responsible citizenship

These ideas evolved over the centuries from educational practices in Europe. The ancient romans wanted young people to receive a practical education that would provide them with

useful knowledge. This perspective is shared by many countries including Kenya. The Romans too argued that students not be harshly punished advocated for more gentler and sensitive approaches

During the middle ages, the influence of early Christian education that was a blend of Greek, Roman and Hebrew ideals was witnessed. The tradition of churches taking responsibility for secular education is traced to this period. Today in Kenya faith based organizations play a pivotal role in curriculum development and implementation.

During the reformation many church leaders believed that the bible was the repository of wisdom and hence desirable for all learners to learn to read so that they might access to its truths. This led to increased demand for biblical and other materials in the vernacular for reading to interpret the bible. The campaigns by Martin Luther King for Protestants and the counter campaigns by Ignatius of Loyola (founder of Jesuit schools) led to increased church activity in elementary, secondary and higher education

Later during the renaissance, there was increased quest for knowledge that led to opening up of education to the masses. This led to the opening of humanistic schools and a growing emphasis on the worth and importance of the individual evolved. Schools concern for meeting individual differences developed from this perspective. An increased degree of enlightenment during the renaissance enhanced communication between the various nations in Europe and beyond. This led for demand in vocational skills related to national and international trade. The school curriculum started to have increasing amount of vocational studies in the form of book- keeping, business arithmetic among others. Demand for higher cadre of professional people with special faculty specializations during this period led to the establishment of universities.

Francis Bacon The  
next major event is the scientific movement in education. In the early 17<sup>c</sup>

(1561

1626) helped to establish the idea that truth could be challenged and modified through observation and careful weighing of evidence. This led to the philosophy of realism. This is the foundation of the scientific method that continues to be tremendously important to today's education. John Amos Comenius (1592-1670) promoted the ideas of organizing learning into sequential, graded schools and of viewing education as something that should prepare people for happy lives.

In the 18 century Johann Heinrich Pestalozzi (1746-1827) suggested that education should take place in a caring atmosphere and should function as an agent to improve society. This view that education could promote social improvement was considered radical in Pestalozzi's time but today it's everyone support the same. Pestalozzi also introduced the idea that teachers require special training.

In the 19<sup>th</sup> century mandatory public schooling had been adopted as public policy in Europe as compared to Kenya where it happens a few years ago and still facing some resistance. An educator Friedrich Froebel (1782-1852) worked hard to convince people that they should support efforts to provide young children with well-planned educational experiences.

In the same 19<sup>th</sup> century another educator Johann Friedrich Herbart (1776-1841) developed a standardized format for lesson planning. The lesson plans you will use probably will include many features initially introduced in Herbart's design. Similarly was a lot of interest in the importance of psychology in learning



### **The pre-independence education commissions and ordinances**

The British government was against the introduction of an academic curriculum in the colonies. They advocated for industrial education that was adapted to the needs of the local people. The

school was to be conducted in accordance with native customs in matters of dress and etiquette, in order that pupils may not become denationalized or consider themselves a class apart, which was meant to combat political agitation.

The missionaries whose objective were evangelism and enlisted Africans to join their churches established schools in the villages and provided simple education mainly reading, writing arithmetic

and religious instruction. Post primary education was not encouraged. This stand was supported by colonial administrators who argued that more emphasis should be given to manual labor

The expansion of missionary education as a result of the railway line led to competition amongst the missionaries. Consequently they taught their pupils to mistrust their rivals leading to animosity between them. They had limited funding, trained teachers were in short supply and no curriculum was available.

In 1909 Nelson Fraser (Fraser Commission) was commissioned to recommend a system of education for the east African protectorate. In his report he placed emphasis on the provision of industrial education, and education facilities for Africans should be provided by missionary societies on the grounds that any education without Christian instruction would lack a moral foundation. It also recommended establishment of the department of Education was founded in 1911 (Sifuna, Chege and Oanda, 2006).

Government –grant –in aids started to support mission schools that which provided technical education.

A number of schools were offering industrial skills such as smithing, carpentry, agriculture and typing.

Racially segregated schools were started for Europeans and Asians.  
[The Phelps-Stokes Commission](#) (1924) recommended creation of Director of education in several colonies, creation of local advisory committee on education and founding of schools to train Jeanes teachers (Berman, 1971). Further they emphasized on the need for Africans to receive technical education and emphasize on agriculture. Following phelps –Stokes commission government came up with an education ordinance to demonstrate its commitment to supervise and direct education at all levels with assistance of three advisory committees to deal with European, Asian and African education.

All schools and teachers were to be registered and the director of education was empowered to inspect all schools. The district education board ordinance of 1934 established District Education boards were



set up to assist in the management of all schools. They handled the allocation of grants-in- aid, fees and scholarships, salary scales as well as maintaining register of schools

- **Post-Independence education commissions and committee**
  
- **i. Kenya First Education Commission (Ominde Report) 1964/65** made the following recommendations/contributions:
  - Abolition of education on racial segregation to promote national unity
  - Came up with pre-independent statements of the national goals of education (educational philosophy)
  - 7-4-2-3 educational system with CPE at class seven replacing KAPE at class eight.
  - Education that had less emphasis of vocational subjects in reaction to predominantly technical education that was being provided to Africans by colonialists in preference to academic subjects to prepare Kenyans take over white collars jobs that were being done by whites
  - English was taken as the language of instruction and the national official language
  
- **ii. National Committee on educational objectives and policies (the Gachathi Report) 1975**
  - Recommended 9 year primary education with 2 years being for vocational/technical education
  
- **iii. Mackay Report (1981)**

- Recommended the establishment of a second technological university
- Recommended 8-4-4 education system
- Recommended a broad curriculum that was more practical and with emphasizes on self-reliance
- **iv. Kamunge Report (1988)**
- Recommended cost sharing in education with government providing teachers' salaries and parents meet the cost of infrastructure and teaching/learning materials
- **v. Koech Report (1992)**
- Recommended the —Totally Integrated Quality Education and Training (TIQET)‖
- **vi. Kenya constitutional 2010**
- Brought about devolution and early childhood education was devolved the county governments
- Values were to be taught in the school curriculum.
- According to the Constitution of Kenya, 2010 it is imperative that the State Department responsible for education develops and incorporates values in to the curricula at all levels of education.

- The values stated in the Constitution include responsibility, respect, excellence, care and compassion, understanding and tolerance, honesty and trustworthiness, trust, and being ethical
- **vii. Odhiambo report (2012) on alignment of the education sector to the constitution 2010**
- Recommended introduction of 2-6-6-3 education system with grades replacing classes and less emphasis on examinations (summative assessment)
- Recommendations of a competency based curriculum (CBC) that is competency based and nurturing of each learners potential
- **viii. SESSIONAL PAPER NO. 1 OF 2019 on A Policy Framework for Reforming Education, Training and Research for Sustainable Development**

This policy document proposes a system of education and training that emphasizes the attainment of competencies and values at all levels (Competency Based Curriculum). Reforming the curriculum will also emphasize national values, integration of science and innovation, and adoption of ICT technologies.



## Philosophical Foundations

Philosophical perspectives represent cluster of values and attitudes that individuals use to evaluate alternative action options and decide which are preferred.

- What is your personal philosophy?
- What is your philosophy of education?

Your philosophical positions help explain your personal reactions to events you confront in your daily life and what you find personally rewarding and satisfying.

Philosophical foundations constitute the values and beliefs that make up the philosophies of life and of education and have a permeating influence on the other foundations.

Philosophy seeks to provide answers to basic problems and to establish coherence in the whole domain of experience. Since one cannot teach everything, there is need to select content from the vast stores of knowledge possessed by each nation and society. There are broadly two

philosophical schools that guide the selection of curriculum content.

- Traditional school
  
- Perennialism
  
- essentialism
  
- Progressive school

Further, we have other philosophical schools with similarity

- Traditional school
  
- Idealism

- Realism
- Pragmatism

### **Philosophical criteria for selection of content**

The main philosophical schools of perennialism, essentialism and progressivism advance three corresponding theories of subject matter:

1. Subject matter should be taught for its own sake.
- Advanced by perennialists. They believe education system should be stable and its purpose steady.
  - They believe that permanence of the curriculum content and experiences is more important than change
  - Subject matter has a value that is inherent in the subject being taught as well as being intrinsic to that subject

Perennialists lay emphasis on the classical subjects and all students should be exposed to them.

Consider core subjects in the curriculum

- Students should be taught certain basic subjects that will acquaint them with world fundamental values and which make them refrain from regarding modern changes as being the most important
- Man's highest attribute is rationality. Teachers should inculcate good taste, infuse ideas, instill wisdom and implant good sense.

- **ii. Subject matter should be taught for use**

- Essentialists maintain that there are certain essentials that each student in the school should know.
- Essentialists take time to re-examine curricular matters, distinguish essentials from non-essentials and re-establishing the authority of the teacher in the classroom
- Essentialist agree with perennialists that:
- Learning involves hard work. There is need for strict discipline in the process of education
- Teacher's role is to mediate between the adult world and the world of the child.

Teachers are especially prepared for the task of guiding the growth of their pupils •

The heart of the educational process is the assimilation of the prescribed subject matter.

Only subject matter that is useful in everyday life should be stressed in school curricula

- Schools should retain traditional methods of mental discipline. Salient principles and concepts of great scientists and philosophers and artists which have relevance to the world today



1. Subject matter is a medium for teaching life processes and skills.

- Progressivists take the pragmatist view that change, not permanence is the essence of reality
- Educationalists must be ready to modify methods and policies in the light of new knowledge and changes in the environment.
- Progressivism is associated with John Dewey (1948) and advances the following principles in regard to the nature of education and curriculum planning and development 🎬 Education should be life itself not a preparation for living.
- Learning should be directly related to the interests of the child. The child needs guidance and direction from teachers and teachers are better equipped to perceive

meaning in the child's activities

- Learning through problem solving should take precedence over the inculcating of subject matter
- Teachers have to advise not direct pupils. Teacher is only important as stage setter, guide and co-ordinator

- The school should encourage cooperation as opposed to competition. Competition is admissible in situations where it promotes selection for stations in life and other opportunities for which qualitative differentials have to be manifest

Other philosophical schools

### **Idealism**

Traditional view influenced by writings from Plato

- Idealism uses deductive reasoning in its quest for answers to current problems
- In idealism there is over-emphasis on the intellectual aspects of life at the expense of the physical aspects
- Idealism stresses the role of education in the transmission of the cultural heritage as handed down through the ages from the past.
- Just like the perennialists the idealist view is that the student copies and emulates his teacher through the exchange of ideas in teacher-controlled discussions

### **Realism**

- Realism believes in the existence of a real world, divorced from the imagination of the perceiver
- The real world exercises a systematic and accurate regulation of its activities which can be observed in the physical universe
- Realists approach the problems of man in life through the inductive method which is data gathered to form a basis for new principles and generalizations
- Realism accepts God as the motive cause of all existence.



- Realism argue that education should induct learners into their culture and help to adjust to the natural order to live in harmony with the universe.
- Students in the school system are expected to recognize natural laws and to react appropriately to them
- Teachers act as guides making children aware of the true nature of the real world

Realism encourages an objective approach to situations as well as active use of the senses in learning

- Realists advocate a study of physical and social sciences which are instrumental to explaining natural phenomena
- In addition mathematics is considered essential in the description of the universe since it is symbolic and accurate

### **Pragmatism**

- Pragmatism is a progressivist position that sees reality as being in a state of flux or constant change
- To establish truth it is necessary to examine all the possible consequences likely to result from putting the idea into action or into practice
- Science is used to provide answers to man's problems
- Pragmatists employ the realist approach in gathering information and facts, and the idealist approach in generalizing about the facts gathered
- According to pragmatists education:
  - Should enable the learner to experience situations in practice
  - It's a means for recreating, controlling and redirecting experience
  - Should help learners to solve their problems and is to be considered as integral part of life
- Systematic sequencing of learning experiences is emphasized by pragmatists

Curriculum should be organized on the basis of the learners interests and subject matter selected should help the learner to solve problems

## **Psychological Foundations of curriculum development**

Psychological foundations of curriculum refer to those insights gained from psychology which have a bearing on the learning process. They subsume the total knowledge that guides the learning process and makes it possible for the teachers implementing the curriculum to make appropriate decisions regarding the classroom behavior of learners

- Psychologists believe that learning experiences have to be introduced to the learner when such exposure is most effective and most beneficial to him
- Psychology as a discipline offers certain principles that govern the process of learning which a teacher who understands them will find helpful in meeting some of the demands in the learning process
- It helps the curriculum designer and the teacher to make decisions in relation to:
  - Sequence-in the stages of development
  - Organization-grouping of learning experiences for optimal effect

- Methodology-dealing with the question of what methods and approaches are likely to promote and guide learning most effectively.

Formulation of appropriate educational goals

- Decisions regarding the scope of the curriculum
- Theories of **human development**, theories of **learning** and the **pedagogical principles** derived from psychological theories must all be considered in **determining the curriculum**.

**Psychology focuses on learning and teaching through three main theories:** Behavioural theories, Cognitive theories and Humanistic theories

- **Behavioral Theories**
- Behavioral psychology is the oldest theory of learning. This group of psychologists have contributed to decision making in curriculum through their findings and theories regarding conditioned responses in the area of emotions.
- Knowledge of this helps in dealing with problems connected with desirable classroom atmosphere.
- Knowledge about the learner is relevant in making curriculum decisions during curriculum planning such as:

ü Selection and arrangement of content ü The choice of learning experiences by which this content will be manipulated and by which the objectives not achievable through content alone can be attained. ü Plans for optimum conditions for learning.

These decisions cannot be made adequately without knowing a good deal about learners and learning.

v Knowledge of how intelligence functions and how capacities develop, enable curriculum developers tailor curriculum content to needs and capacities of children v Knowledge about transfer of learning help in making decisions about the efficiency of learning

i.e., how to make whatever is learnt in school most useful for the rest of life and how to apply it to things other than those in which the school experiences centre.

v Knowledge about the total development of the individual can help discover what the total range of objectives can be and how the curriculum can accommodate or develop this range.

- Application of Behavioural Psychology is illustrated by such learning trends as:
  - o Micro-teaching, Instructional training models, Individual learning, Direct instruction and **Mastery learning**

- **Cognitive Theories**

- **Most learning theories today are cognition-oriented (Intellectual).**

**Jean Piaget**, a Swiss Psychologist, formulated theories of learning, language and mode of thought of the child. He observed and recorded in minute detail the spontaneous reactions of many children to a wide variety of situations

He identified **four stages of cognitive development** as:

1. i.                   **Sensory-Motor stage (0-2 years) (Infant stage)**

- No abstract reasoning
- Thoughts of infants are intuitive, closely linked with actual physical action and immediate observation.
- The infant cannot imagine of consequences of action unless he/she actually carries it out.
- The child is unable to draw logical conclusions from his activities and experiences.

ii.                   **Pre-operational or pre-conceptual stage (2-7 years)**

- The child manipulates things mentally but only objects/materials that are concrete.
- Mental operations develop with active exploration of things in his immediate environment
- The child can appreciate a variety of living things and objects around him
- He/she develops ability to discuss and communicate,
- can recognize shapes and patterns, and
- Can group things together according to a given criteria e.g., shape, colour, living or non-living etc.

- **iii. Concrete operations (7-11years)**

- The child performs varied, more powerful mental operations,
- Problems are solved in a more ordered and quantitative way
- can visualize objects from different angles  Can visualize the shape of cross- sections.
- He can classify living things and non-living materials in different ways

- **iv. Formal operations or abstract thinking (over 11 years)**

- The child can think about abstractions
- He is no longer tied to the concrete and to the here and now  can use hypothetical reasoning.
- Abstract logic and mathematics are now at his disposal
- He is able to distinguish observations which are pertinent to a solution from those which are not
- He is able to undertake controlled experimentation

All children pass through Piaget's stages of cognitive development in the same order but at a different rate. The age at which a child appears varies depending on:

- Child's intelligence
- Previous learning experience



- Social and cultural background
- **Implications for the curriculum v Readiness-**
  - Each stage of development builds on an earlier one and forms the foundation for the next stage
  - Concepts are acquired in sequential hierarchies and therefore certain concepts are fundamental for acquisition of those that follow
  - When planning the curriculum we must take these stages of development of the learner into account
  - The planning of the curriculum should be considered from the logic of the child rather than the logic of the subject matter
  - Should not hurry the syllabus coverage until the earlier cognitive stages have been mastered
  - Time is the servant not the master of the child
  - You cannot force the child to develop understanding faster than his absorption of the related experiences, he will learn when he is ready i.e. when he has reached the appropriate stage of intellectual development
  - Teachers can stimulate children into readiness-the learning process can be speeded up or slowed down by appropriate or inappropriate learning situations v Sequential **learning-**
    - This involves awareness of the stages of development and learning hierarchies through which the learner must pass as he/she advances from one concept to a more sophisticated one and the corresponding sequencing of teaching /learning experiences and the materials and resources used.
    - The task of the teacher is to figure out what the learner already knows and how he reasons so as to ask the right question at the right time so that the learner can build on his own knowledge v

**The need in general to pass from concrete to the abstract;** concrete experiences first, the

word later v The process of forming a concept takes longer than had been previously thought

v One must cross the line between ignorance and insight many times before real understanding comes v Within each subject or discipline there are progressively more complex concepts v Childs intellectual development is enhanced by parental and environmental stimulation

- Parental stimulation-children who receive affection and care develop more rapidly intellectually than those to whom proper parental attention is denied
- Cultural stimulation- Lack of toys/picture books and other stimuli can retard development of manipulative skills
- Use of pictorial material –lack of toys can retard cognitive development.

v Health and nutrition-lack of proteins and vitamins can retard intellectual growth

- Humanistic Theories

**Humanistic approach to curriculum is the most recent learning theory.**

Main proponents include:

- Maslow's on Human needs
- Roger's on Freedom to learn
- Guilford's on Structure of intellect

- Dewey's on Reflective thinking
- Stenberg's on Critical thinking
- Brunner's on the structure of subject and inquiry/discovery methods
- Gestalt's theory on shape, form and configuration (stimulus vs response)
- In this theory curriculum is after the process, not the product, focuses on personal needs, not on subject matter, and clarifying psychological situations.
- It emphasizes that curriculum should be based on societal needs. This approach include:
  - o Formation of desired attitudes, Development of humane feelings, Self-Actualization, Freedom to learn and Value Classification ü

#### Maslow's on Human needs

- Basic needs
- Physiological needs (food, water, warmth, rest, sex)
- Safety needs (security, safety)
- Psychological needs
- Belongingness and love needs (intimate relationships, friends)
- Esteem needs (prestige, and feeling of accomplishment)

- Self -fulfillment needs
- Self- actualization(achieving ones full potential , including creative activities)
- You should note that behaviourist component is need for planning and the curriculum. **Cognitive component** is for developing the intellect through subject content and **humanist component** is for instruction
- **Application of Knowledge of Psychological Foundations to Curriculum Development**
- Curriculum developers/teachers can apply knowledge of certain basic psychological principles in:
  - Meaningful selection and guidance of experiences in order to contribute to maximum development of the learner
  - Organizing and implementing the curriculum in relation to varying rates of mental growth
  - Providing flexible curriculum to allow for diversity of character and learning readiness among the learners

An understanding of certain basic psychological principles will enhance the effectiveness of planning, design and development of the curriculum in a number of ways:

- The physical, health and physiological status of the learner in the classroom has a bearing on the rate at which he will learn
- A learners nervous condition also has an effect on his mental readiness
- Curriculum should be developed in accordance to different age groups, corresponding to different age-grade level
- Different age groups have unique challenges which require that curriculum is planned

taking into account the common challenges of all youth and the specific challenges in the lives of individual students

- Learners have their own interests and aspirations. these interests and aspirations are important determinants of the curriculum structure and content and thus influence learning effectiveness
- Effects of rewards and punishment on the process of learning. Children repond better to learning when teachers put emphasis on rewards than punishment-teacher should plan rewards in organizing learning experiences
- Learning environments should meet learners basic, psychological and self-fulfillment needs...permission to attend to call of nature, provision of water, food, clean toilets, adequate lighting/ventilation, avoid ridicule and demeaning punishments etc will contribute to suitable learning environment.
- Importance of parental and environmental stimulation
- Child rearing practices can have an important effect on intellectual development. Children on whom care and affection are bestowed develop more rapidly

intellectually than those to whom proper parental attention is denied



- Cultural stimulation
- The degree to which a child's intellectual potential is developed depends to a large extent on the stimulus he receives
- Lack of toys and other such stimulus can retard the development of manipulative skills
- Health and nutrition
- Lack of proteins and vitamins can seriously retard intellectual growth. For example Kwashiorkor which is caused by lack of protein in the diet causes mental apathy and slowness and feebleness of movement.
- The cognitive development of children is retarded ; they lose normal curiosity and desire for exploration that is natural to the child and seem to show no interest in their surrounding

### **Implications of psychological foundations for selection of content**

- In selection of suitable subject matter and appropriate teaching/learning experiences for the curriculum:
- consider interests and needs of students,
- special characteristics of children at different ages,

- stages of human development
- Factors such as stimulation, maturation, motivation which bear on intellectual development
- Nature of learning process
- How knowledge can best be sequenced and organized for efficient learning and teaching
- New ways of knowing and of teaching
- Understanding of the academic disciplines and the structure of learning and teaching



## **Sociological foundations of curriculum**

Knowledge deemed relevant to the curriculum will relate to relevant contemporary issues of society and of the world at large. A curriculum has to guide and orient pupils towards the culture in which they will live their lives.

This will embrace the customs, values, beliefs, techniques, institutions and patterns of social living of the society.

A curriculum is a cultural selection and a child must be taught

- The accepted values and mores of his society
- General knowledge and the attitudes appropriate to an educated person
- A particular skill, so that he can earn his living and make a contribution to society.

In 1957 Smith, Stanley and Shores advocated a —common culture core curriculum — in terms of broad social problems and themes of social living for the United States of America. Dennis Lawton (1973) argues for selection from culture of what seems most significant and from body of knowledge and experiences. Such a selection leads him to five cores of the curriculum, corresponding to Professor Paul Hirst ‘forms of knowledge’. The latter argues that one of the primary aims of education is the development of a rational mind. He says knowledge consists of forms of thought with their distinctive and characteristic logical structures called disciplines or subjects. The five areas or core are:

- Mathematics
- Physical and biological science
- Humanities and social studies (history, geography, classics, literature , religion)
- Expressive Arts (music, drama, physical education, painting, pottery)
- Moral education

Often not sufficient attention is given to including the culture, the traditions, of the home, of society, in what goes on in the schools



Sociologists use three different perspectives to address the functioning and performance of institutions such as education.

These perspectives are:

#### Functionalist perspective

- This perspective analyses the role of the school in maintaining an orderly and efficient society
- Functionalists contend that if all institutions are functioning well, then society works well
- Functional analysis focuses on the dominant values in society and emphasizes the benefits of a common language, common values and development of useful skills in order to assimilate diverse groups and mold them into a functioning and cohesive society
- Functionalism is a dominant perspective for viewing and evaluating education. When there are dysfunctions in society, the schools are quickly viewed as both one of the causes of the problem as well as part of the solution
- The functionalist perspective provides some guidance for addressing persistent education problems e.g. school failure where they suggest that the school should work in conjunction with the family
- Functionalism thus influences education policy and suggestions for reform. The business and the industry groups issue reports stating that the schools should be doing more to prepare students for the world of work
- What current school practices can you identify that are consistent with the functionalist perspective?
- Conflict perspective

- Conflict perspective begins with the contention that power relationships and conflicting interests influence education policy and practice
- Schools are viewed as places where power groups compete for advantage.

It is in the interest of the dominant power and status groups to control and shape education in order to maintain their advantage

- Since control of education is basically in the hands of those in power, schools are places that represent the power groups in society and therefore perpetuates social inequality
- Analysis of schools from conflict perspective reveals that schools serving minority groups and the poor tend to be poorly financed and inadequately staffed
- People who use conflict perspective to evaluate education contend that in general, the school has been structured to maintain the dominance of the groups who have the most power and who benefit the most from the system e.g. most people do not realise that merit in schools is usually defined in terms of middle class and upper-class expectations.
- Students from minority groups and from lower socioeconomic status groups must adapt to different set of norms and expectations and thus put them at a disadvantage
- Conflict perspective provides an important lens through which we can view the educational reform movement and proposals for change.
- They propose school programs should address inequality and promote social justice
- Proposals for change should be from perspectives of all members of the society, not just from the those in power.
- We need to ask hard questions such as who is proposing this change. Why are they making these recommendations? Who will benefit? Who will be harmed?
- What is your reaction to the conflict perspective as a tool for analyzing and evaluating schools?
- Symbolic interactionist perspective

- The functionalist and conflict perspectives look at education at a 'macro level'. These perspectives view education as an institution and how it relates to other institutions. Symbolic interactionists contend that these macro approaches miss the dimension of everyday life that shape student futures.
- The focus of symbolic interactionism is 'micro level' or what actually takes place in the classroom.
- They claim students develop their identities, learn their self-concepts, and develop their hopes and aspirations through interactions with others.
- We learn who we are through the mirror of others. Therefore, looking at the interactions between students and between teachers and students is critical in understanding the role of the school in society.
- Symbolic interactionists investigate the formal and informal processes that take place in the classroom (the culture of the school and classroom).
- This involves power relationships between the teachers and the students, the rules that govern classroom interactions, peer relationships and friendship patterns and how all these relate to social class and gender.
- They are interested in who is rewarded, what is rewarded, and what students are learning through their interactions with others in the school environment.
- They investigate the relationship of these dimensions to ethnicity, race/tribe, gender and socioeconomic status.

### **Implications**

- One dimension of interactionism is that of teacher expectations and how those influence student achievement. Research has shown teacher expectations influence the interaction between

teacher and student and this may result in a ‘self-fulfilling prophecy’ so that students behave and achieve in a manner that is consistent with teacher expectations

- Another aspect of classroom interaction focuses on the cultural background and gender of the teacher and the students. E.g. students from cultural backgrounds different than the teacher may exhibit behaviours that are normal for them but are misinterpreted by the teacher so that the teacher concludes that the students are rebellious, disrespectful, or unmotivated
- Many current education practices reflect the symbolic interactionism perspective.
- Abolitions of education that was racial after independence and replacing it with a single curriculum that allows all children regardless of tribe, ethnicity, religious belief, social

classes, gender and background was informed by this perspective

- Mainstreaming of special education students into the regular classroom. This was based on the interactionist notion that placing special education students in regular classroom would promote interaction between students that would change the expectations and perceptions of everyone

### **Social forces that have had major influence on schools and curriculum**

The aims/goals and objectives of education are determined by the culture of its society. The influence of social foundations on the curriculum can be considered in terms of:

#### **1. Purposes of the curriculum**

Schools serve as agents for social growth, development and reform. They perform the following functions

- Preservation and transmission of cultural heritage

- Transformation of culture. Education should help the youth to adjust to changes in society and develop ways of dealing with problems
- An instrument for individual development. The school should help the youth to
  - Realize their potential
  - Determine their individual goals
  - Attain their individual goals
- Pressures influencing the curriculum
  - The public in general and various social forces influences the curriculum. Pressure groups influence the nature and direction of the curriculum
  - Society uses both formal and informal means to put pressure on the school system which in turn influences curriculum development
  - Characteristics of the students experiencing the curriculum

Students/learners are prime consideration in [curriculum design](#) and development. Students are unique individuals and this has the following implications

- The social structure of which a student is a part dictates the relevance of the curriculum. Teachers should relate the curriculum to the local surrounding and also achieve the set objectives of the curriculum in the subject area
- Aspirations of students. Curriculum should help learners make correct choices along their lines of aspiration. It should also help build positive attitudes in the student and to raise the level and quality of their aspirations
- Effect of environment on mental development. Schools and teachers should create suitable learning environment to help learners deprived of such to improve in their academic performance.

### **The role of other socializing agents**

The family and religious institutions are two important institutions that have great influence on the school curriculum

#### **1. The family**

Knowledge of the family background helps the curriculum developer and the teacher in the selection of appropriate learning experiences

#### **1. Society around the school**

The society around the school affects the process of curriculum planning

There should be provision for interaction between the school, the society and the family to enrich the school curriculum



Religious institutions

Religious education is part of the school curriculum at both primary and secondary school levels.

Religious education develops an appreciation in the learner and sense of self-respect and respect for life as well as the contribution of such study towards the development of the quality of life.

# CURRICULUM DESIGN



## Lesson Learning Outcomes

- Explain the term curriculum design
- Discuss models of curriculum design
- Discuss the criteria for selection of content/learning experiences
- Discuss the principles for organizing content and learning experience
- Discuss patterns of Curriculum design
- Explain the components KICD designs in curriculum



## The concept Curriculum design

According to Taba (1962) **curriculum design** is a statement which identifies the elements of the curriculum, states what their relationships are to each other, indicates the principles of organization and the requirements of the organization for the administrative conditions under which it is to operate. Curriculum design is a more complex process that results from a particular model of curriculum development. Curriculum design is about what we teach, why we teach it, and how we teach it. It is an

ever changing kaleidoscope of changing, shifting positions. If curriculum is defined more broadly than syllabus or course of study then it needs to contain more than mere statements of content to be studied. A curriculum has at least four important elements: content; teaching and learning strategies; assessment processes; and evaluation processes.

The process of defining and organising these elements into a logical pattern is known as curriculum design. Curriculum writers have tried to place some order or rationality on the process of designing a curriculum by advocating models.

In designing a curriculum we make appropriate choice of objectives, content and learning experiences sequence them and plan for their evaluation. However the process of curriculum planning is much wider and involves:

- Surveying the current curriculum and its context (situational analysis)
- Setting goals
- Deciding a strategy for change
- Designing and developing a new curriculum to meet these goals in that context 🎬

Implementing it and following it up



## Models of Curriculum Design

We are going to look at two common models of curriculum design, namely Tyler (1949) model and Taba (1962) model

- **1. The Tyler Model**

The basic four step model comes from Ralph Tyler, in 1950, with his famous four questions:

- "What educational purposes should the school seek to attain?"
  - What educational experiences can be provided that are likely to attain these purposes?
  - How can these educational experiences be effectively organized?
  - How can we determine whether these purposes are being attained?"
- These questions are easily turned into four steps:
- Purposes (objectives),
  - Selection of learning experiences,
  - Organization of learning experiences and

- Evaluation.

Step one is the most important in this model because all the other steps are derived from it. Tyler said any educational programme should be guided by objectives. Because of this emphasis on the objectives the model is referred to as the objective model of curriculum development.

Tyler identifies three sources of educational objectives

- Study of the learners-educational objectives represent the kinds of changes in behavior that the school seeks to bring about in pupils
- Study of contemporary life in society-the school is supposed to prepare the youth for life in the society
- Suggestions from subject specialists-this is the source commonly used where subjects still form the main basis of the curriculum. Since we will depend on the subject matter to design the curriculum, the expertise of the subject specialist is critical

The objectives arrived at from the three sources should then be filtered through philosophical and psychological —screens— in order to arrive at the final refined objectives.

In step two the objectives are used to generate certain learning experiences and content. Tyler insists that not just facts and concepts we learn but skills and processes we learn. This content must be limited to manageable scope and sequence. The nature of knowledge, the needs of the learner and of society as well as certain other considerations enable us to decide on the scope and sequence. He outlines types of learning experiences and suggests the principles for selecting such experiences.

Step three Tyler defines what he means by organization and provides criteria, principles and structures for organizing learning experiences

In step four explains that evaluation is the process for determining to what extent educational objectives have been achieved by the programme of curriculum and instruction. Evaluation is divided into two types: formative and summative.

Tyler model is celebrated as the first model for curriculum development and all other models have been developed using it as a prototype. However several criticisms have been levelled against Tyler model. First, it is too simplistic and does not make adequate allowance for interrelatedness of the separate parts (its linear). Secondly the model does not mention explicitly which it's assumed under the learning experiences

## **1. 2. Hilda Taba Model**

Taba (1962) asserted that a curriculum must contain certain basic elements namely; aims and objectives, content, learning experiences and evaluation. She further said that these elements must be selected on the basis of certain considerations like principles of learning, nature of the learner and knowledge.

The model is similar to the Tyler model as Taba too emphasizes the importance of objectives. However

Taba adds content as element in her model. She also emphasizes on the interrelatedness of the elements. The Taba model has seven steps which are cyclic namely,

- Diagnosis of needs
- Formulation of objectives
- Selection of content
- Organization of content

- Selection of learning experiences
- Organization of learning experiences
- Evaluation

- **Backward Design Model**

There are three approaches to curriculum design, namely:

- Forward design/input approach. It focuses on the inputs first. I.e input-process-output
- Central design/process approach. It focuses on the process first followed by input and output
- **Backward design/output approach/outcome approach.** It focuses on the outcome first

then process and input follows. Backward design is also called backward planning/ backward mapping/ output approach/outcome approach. The CBC adopts backward design model where the curriculum focus is on the learning outcomes and not the content like with the previous curriculum designs. Backward design aligns everything in the classroom towards final goal, keeps students interest, allow for differentiated approaches in activities and meets diverse needs of students. The design has three stages namely:

- Identifying the desired outcome
- determining the assessment

- planning learning experiences and instruction.



## Criteria for selecting learning experiences

### Conformity with objectives

The experiences selected must give the students the opportunity to practice the kind of behaviour that is implied by at least one or more objectives.

- **Validity**

The learning experiences-content, learning activities or materials- must be authentic or consistent with development in the subject or knowledge area. Teachers need constant in- servicing and self education to keep abreast of changes in their disciplines to avoid use of textbooks and methods that are out of date.

- **Significance**

Learning experiences must be meaningful-must contribute to the process of learning. Is the content logically central enough to apply to a wide range of problems? The objectives should guide the teacher so as to cover content that would be understood by learners of a given level. Coverage of too much content and activities may confuse the learners if done without taking care of breadth and depth.

- **Learner-ability:**

Learning experiences must be appropriate to the learners-must be adjusted to the abilities of the learners. Consider the ability of the learners and their individual differences. Learning also depends on circumstances prevailing in the schools, classrooms and society at large. The reactions desired in the experiences must be within the range of possibility to their present attainments and predispositions-the learning experience must involve the kind of behaviour that a child is able to manifest.

- **Interest**

Learning experiences must be inviting so that the learners are stimulated to hard-work.

Learners will be motivated to perform interesting activities thus enhancing the rate at which learning take place. Selection of learning experiences should not be done solely on the basis of pupils interest unless the experiences address already set objectives.

- **Satisfaction**

The learning experiences must be such that the pupils derive satisfaction from carrying out the kind of behaviour implied by the objectives. If the experiences are unsatisfying, the desired learning is not likely to occur-there must be satisfaction obtained in the identified behaviour for the learning experience to become effective.

#### **Vii) Relevance**

The learning experiences must be applicable to life situations now and in the future. Although relevance is embedded in the objectives, it is important to look at it in immediate terms. The learner must see his school experience as being related to his life outside the school. The experience should be based on or reinforce what he has experienced at home. The learner should be able to repeat that experience to solve his problems and problems of society in real life. If what learners experience is similar to what they have experienced before or is related to a problem they see in real life , they become very curious, attentive and eager to have a go at it.

- **Consistency with social realities.**

A learning experience may address the objectives but contradict the actual situation at home or in society. Change, rapid change, is characteristic of the world we life in. The material and experiences



selected must help develop minds which can cope with change. Often what is selected is based too exclusively on what has stood the test of time.

- **Economy/Comprehensiveness**

Do the content and learning experiences provide for a wide range of educational objectives? There are certain individual learning experiences that are likely to bring about a number of learning outcomes. Learning experiences that lead to the attainment of more than one objective permits economy in the use of time-these are preferable to those that can only achieve a single objective.

- **Reinforcement.**

Those experiences that reinforce previous or other experiences in order to achieve objectives can be very useful. Certain Objectives, by their nature, cannot be achieved well through a single experience there is need, therefore, to understand objectives and the corresponding learning experiences so as to be able to decide on and select experiences where reinforcement occurs



## **Principles for Organizing Curriculum Content**

The selection and organization of content/learning experiences can be problematic. Curriculum design must take into account the following principles

1. **Scope**-This relates to the breadth and depth of content.

- What content from the disciplines and non-disciplines should be included in the curriculum?

- What content should be considered as electives?
- How much content can be learned in one subject area in one year or in a cycle, in a topic or theme; in a scheme of work; in a single lesson?

It is generally agreed that the scope of curricula content is regulated partly by the goals and objectives generated during the diagnostic stage in curriculum planning. These objectives also help to regulate curriculum objectives

- **2. Sequence-** Sequence deals with order of content and experiences. It addresses the problem of ordering the curricula offerings so as to optimize learning.
- **Content can be organised and sequenced by drawing on its substantive structure** i.e., the logic of the discipline. Here the content is arranged on the basis of key concepts and the order of learning is determined by what concepts are pre-requisite to the learning and comprehending of other concepts. The logical arrangement of curricula content depends entirely on the nature and structure of particular subjects.
- Sequence based on psychological principles draws on our understanding of growth, development and learning. Piaget (1960) provides a fairly useful framework for sequencing learning experiences based on the knowledge about how individuals function at various cognitive levels of development.

The learner needs to acquire a certain level of development physically, mentally, socially and morally to be able to deal with certain concepts and situations

- **Chronology-** this is a rather silent organizer for sequencing curricula content. Subjects like History and Government are organized chronologically. The assumption is that historically events take place successively in time and that prior happenings lead to the present events

- **Continuity-** this refers to the many occasions which individuals will experience content at various levels with an education system. It is the vertical reiteration of curriculum elements affecting the occasions on which certain experiences will be scheduled during consecutive periods of time. Curriculum specialists often extend themes vertically throughout an entire educational system. Continuity deals with continued presence of curriculum demands like concepts or issues and it is necessary to ensure that there is a recurring and continuing opportunity for a skill to be practiced and developed.
- **Integration-** this emphasizes the horizontal relationships between various curricula areas in an attempt to interrelate content, or learning experiences in order to enable the students to perceive unity of knowledge. Curriculum experts generally feel that learning is more effective when content from one field is linked meaningfully to content in another

field. Integration is a reaction to the traditional emphasis of separate subjects in the curriculum. It is generally used in the broad fields and correlated curriculum designs. It has several advantages:

It assists students in creating a comprehensive understanding of the content experienced in an educational setting

- Helps the student to get an unified view and to unify his behavior in relation to the elements dealt with
- Helps to create unity in the student's outlook, skills and attitudes.
- Balance- a balanced curriculum implies structure and order in its scope and leads to the achievement of educational objectives or order. A balanced curriculum means a short program which adequately ensures that each educational objective is achieved. A balanced curriculum should develop all major areas of human competence like communication, problem solving, understanding concepts and environment, health care, recognition of interests, and deep interest and skill in aesthetics. In addition to individual development, a balanced curriculum should address all the national goals of education.

Problems of curriculum balance:

- There is a dilemma as to how much of the cultural disciplines students should learn and at the same time cater for their individual needs and interests-how to strike a balance between societal expectations and the needs of individual learner in the school context
- Science subjects versus the arts and the humanities-whether to emphasize science or balance it with arts.
- Technical and vocational subjects versus academic subjects-emphasis on practical and practical curriculum-dilemma is as to what to include in the curriculum given the limited time available for learning at school

The rural urban dichotomy in Kenya-strike a balance between needs of the urban and rural folk should we have different curricula-one urban and another rural.



## Patterns of Curriculum Design

Formulation of objectives and selection of learning experiences is followed by organization of these experiences for effective presentation and learning. Patterns of curriculum organization also referred to as patterns of curriculum design refer to the main structural elements in which the learning experiences are to be organized

Patterns of curriculum design can be divided into three main categories, namely, subject- centered designs, learner-centered designs and learning experiences organized in structures of sequential courses like Biology I, Biology II, Biology III, etc.,

### **Subject-centered curriculum designs**

- **The traditional subject designs**
- The oldest and the most popular form of curriculum organization
- The subject design organizes the curriculum in different and specific subjects each of which represents a specialized and homogeneous body of knowledge-content.
- Emphasis is placed on separate academic areas called subjects/areas with their own logical order and sequence as decided by subject specialists

### **Advantages of subject designs**

Most systematic and effective organization for acquainting youth with the essentials of curriculum heritage

- Subjects constitute a logical and effective method of organizing and interpreting learning students learn most economically and efficiently
- Because of the long established tradition teachers and parents tend to support what they are familiar with
- Most teachers have been trained in subject curriculum-they understand it better
- Curriculum planning, implementation and evaluation is easier in the subject centered curriculum
- The full intellectual powers be best developed through the study of logical organization of subject matter
- Textbooks and other curriculum packages on the education market tend to be organized to fit the established subject categories-according to subjects

#### **Disadvantages of subject centered curriculum designs**

- Lack of integration of curriculum content-compartmentalization and atomization of knowledge leading to fragmentation in the students understanding (the learner fails to see the relationship between subjects and yet naturally knowledge exists as an integrated whole)
- Subjects tend to be detached from the reality of students-the design is not based on a functional use of knowledge and is detached from the concerns and the events of real world
- It focuses on the intellectual performance (content) at the expense of other aspects of human development

It tends to relegate teaching to knowledge and recall of subject matter-recitation, rote and passive learning dominate the learning process and the teacher is the centre of attention or activity

- It leans heavily on verbal activities and is too much oriented towards examinations which tests memorization rather than understanding

- **Correlation**

**Correlation** tries to reduce fragmentation of curricula content by drawing from various subjects or disciplines in learning a selected theme, e.g., content can be drawn from

- Economics, history and religion to teach the theme of ‘war and peace’
- Geography, history, economics, agriculture and civics to teach ‘patterns of settlement in independent Kenya’

Though correlation does not the fragmentation of the subjects (they still remain separate) it tries to show the learners the relationship between the various subjects

- **Broad fields**

Also commonly referred to as **fused design** is an advanced stage of correlation in the modification of subject matter curriculum. It involves a merger of different but related subjects into new course to overcome fragmentation e.g. ., ‘social studies’ resulting from the merger of civics, history, geography, sociology, anthropology, social psychology, economics, social ethics and other subjects that deal with man and his relationships in a given physical environment. **The Gachathi commission (1976) had suggested five broad areas**

- Languages (English and Kiswahili)

Mathematics (arithmetic, algebra and geometry)

- Sciences (biology, agriculture, health education, physics and chemistry)
- Culture studies ( history, civics, music, art and craft, physical education, and social ethics)
- Pre-vocational studies (woodwork, masonry, business education, metal work etc

**Philip Phoenix's combination suggests six different clusters of knowledge**

- Symbolic: ordinary language, mathematics
- Empirics: physical sciences, biology, psychology. Social sciences
- Aesthetics: music, the visual arts, arts of movement, literature
- Synnoetics: personal knowledge and interpersonal relation ships
- Ethics: moral knowledge
- Synoptic: history, religion, philosophy **Advantages**

• This design is more evident in primary schools. However it is becoming more relevant to high school and higher education as demand for education that is relevant to the current and anticipated needs of learners' increases.

- It also allows meaningful integration of related content.
- It can meet students' interest and furnish with diverse experiences

**Disadvantages of broad fields**

- Content so organized can be superficial and lack depth of treatment



Content structuring can still be fragmented and fail to relate to the actual world of the learner

- **Core-curriculum**

Core curriculum came as a reaction against the fragmentation and piecemeal learning's associated with separate subjects. **Core-curriculum** means that part of the school curriculum that is essential and a common requirement for all students. Other terms associated with core- curriculum include:

v Common

learning's v Unified

studies v Basic education

The core-curriculum draws on social issues, moral content and values and help to: vDevelop integration of knowledge to serve the needs of the students vPromote active learning and a relationship between life and learning vServe the needs of society vGive students the opportunity to make choices for the subject area of study through electives

**Advantages of core-curriculum**

- Emphasizes a core of social values
- It combines subject matter with the realities and challenges of the day
- It aims to integrate learning by unifying subject matter for studying social issues and problem areas related to individuals as well as society
- It takes care of the needs and interests of the learner

It is socially rather than cognitively oriented

**Disadvantages of core-curriculum**

- Fails to offer significant and systematic knowledge. It also poses organizational problems in schools
- Calls for variations which modern teachers cannot cope with because they lack the broad competence and training required to arrange content around focuses or issues
- It requires teachers of varied specialties, an adequate curriculum guide and materials which are usually not available or in short supply

### **Learner-centered designs**

These are designs where the child or the learner is the centre of the curriculum process and his active participation in the learning process is the main focus of these designs. It is associated with the philosophy of Jean Jacques Rousseau and works of John Dewey with his progressive movement in the US.

**Characteristics of learner centered designs** v They take their organizational cues from individual students rather than from content- the learner is the centre and end of school activities v They focus on the learners' needs, interests and experiences v They are highly flexible, with many options available for learners v Ideally the learners are supposed to plan their own curriculum, usually with various degrees of teacher input v Learners are instructed individually at appropriate points

- **Activity curriculum**

Originated in Europe and is associated with works of Rousseau and Pestalozzi but gained prominence during the progressive movement in 1920s and 1930s. Kilpatrick emphasized purposeful activity based on the child's experiences, needs and interests. He claimed that desirable learning would be produced by active participation of the individual in his environment. From the activity movement there emerged any pedagogical concepts that are associated with activity learning:

v Life experience units v Projects v Social enterprises v

Field trips v Centers of interest

All the activities involve; v A section of problem

solving v Active participation by students v

Socialization of the student in the world around him v

Greater school community linkages

- **Social and life designs**

The design is based on social processes and life functions. It is concerned in promoting life that is needed in the society and providing solutions to prevailing social problem. It attempts to provide a relationship between the content of the curriculum and life. Stratemeyer, Forkner and Mckim (1957) suggested a design organized around persistent life situations:

- Situations calling for growth and individual capabilities
- Health
- Intellectual capacity
- Moral choices
- Aesthetic expression and appreciation

- Situations calling for growth in social participation
- Person-person relationships
- Group membership
- Situations calling for growth in capability for processing environmental factors and forces such as:
  - Natural phenomena
  - Technological phenomena
  - Economic-social, political structures and dynamics

We may look at curriculum designs either in isolation or some kind of relationship. In most school curricula both the traditional and the new subjects form the basic structures. However, in the process of selecting the subject matter and in [curriculum implementation](#), several ideas from the modified subject designs and from the largely child-centered designs have been incorporated. To develop any curriculum we must conceptualize a framework that will guide our own design which will in turn guide the process of curriculum planning and development

# CURRICULUM DEVELOPMENT PROCESS

## Introduction

In this lesson we explain the term curriculum development and curriculum development process, then discuss the stages in the curriculum development process. The role of the Kenya Institute of Curriculum Development (KICD) will then be outlined. Approached to curriculum development will then be discussed and finally the role of various Agencies and Institutions involved in curriculum development will be explained.

## Lesson Learning Outcomes

By the end of this lesson, you will be able to:

- Discuss the stages in curriculum development process
- Outline the role of the Kenya Institute of Curriculum Development in curriculum development (including its structure and functions)
- Explain the role of various agencies and institutions in curriculum development process.

## : Meaning of curriculum development and curriculum development process

**Curriculum development** is the planning of learning opportunities intended to bring about certain changes in the learners and the assessment of the extent to which these changes have taken place.

Curriculum development entails the writing, trial and testing of instructional plans and educational materials. The curriculum development process systematically organizes what will be taught, why it will be taught, who will be taught, and how it will be taught

Most models of curriculum development describe/portray/depict it as continuous process, with no single fixed starting point.

Stages in the curriculum development process:

- Needs assessment
- Policy formulation
- [Curriculum design](#) and syllabus development
- Approval of the curriculum and syllabus
- Development and approval of curriculum support materials
- Capacity building and orientation of curriculum implementers
- Piloting of the curriculum
- Regional implementation of the curriculum
- Monitoring and evaluation of the curriculum

## • : The structure and functions of Kenya Institute of Curriculum development

- **Structure**
- KICD is a semi-autonomous government institution under the Ministry of Education. The institute is managed by a governing council appointed by the minister of education to guide policy matters and activities of the institute on behalf of the ministry.
- The council is made up of senior educational administrators and representatives of religious organisations and educational institutions (Principal secretary state department of basic education, Director Quality Assurance and Standards, CEO Kenya national examinations council, CEO Teachers Service Commission, etc. The chairperson is appointed by the Cabinet Secretary MOE and the CEO of the institute is the secretary.
- The council has two major Committees, namely, **the academic board** and the **finance and general purpose committee**.
- The Academic board guides the professional activities of the institute such as research, evaluation and the actual development of the instructional programmes
- The finance and the general purpose committee guide the executive and administrative functions of the institute. It also controls the appointment and tender committees. It provides support services to the main activities of the institute which fall under the academic board.
- Under the academic board we have curriculum development panels and research evaluation division that carry out professional work. There is a course panel for each level or area of education (course panel for Primary education, Course Panel for secondary education, course

panel for preschool education, course panel for non-formal and continuing education. Course panels also deal with teacher education matters for that level of education.

- There are several subject panels under each course panel.
- Each subject or curriculum area has a panel.
- Each of the subject panels takes care of the related teacher education, research and evaluation.
- Subject panels keep under constant review the existing syllabuses (curriculum designs for CBC), examinations and textbooks in the subject and make recommendations to schools and colleges.
- They also initiate and guide appropriate curriculum projects at the institute
- Subject panels are answerable to the course panel in all their professional activities
- They make recommendations to the course panel which they may approve, make modifications or reject them altogether.
- The academic board must approve all recommendations and programmes before they can go to schools for implementation.
- **Functions**

The Kenya Institute of Curriculum Development is mandated to perform some of the following functions:

- Advises the Government on matters pertaining to curriculum development
- Evaluate, vet and approve, for application in Kenya, any local and foreign curricula and curriculum support materials in relation to the levels of education and training other than university



- Implement the policies relating to curriculum development in basic and tertiary education and training;
- Develop, review and approve programs, curricula and curriculum support materials that meet international standards for—
- Early childhood care, development and education;
- Pre-primary education;
- Primary education;
- Secondary education;
- Adult, continuing and non-formal education;
- Teacher education and training;
- Special needs education; and
- Technical and vocational education and training.
- Initiate and conduct research to inform curriculum policies, review and development.
- Collect document and catalogue information on curricula, curriculum support materials and innovations to create a data bank and disseminate the information to educational institutions, learners and other relevant organizations
- Print, publish and disseminate information relating to curricula for basic and tertiary education and training
- Collaborate with other individuals and institutions in organizing and conducting

professional development programs for teachers, teacher trainers, quality assurance and

standards officers and other officers involved in education and training on curriculum programs and materials

- Develop disseminate and transmit programs and curriculum support materials through mass media, electronic learning, distance learning and any other mode of delivering education and training programs and materials
- Promote equity and access to quality curricula and curriculum support materials
- Promote appropriate utilization of technology to enhance innovations and achievement of a knowledge based economy
- Offer consultancy services in basic and tertiary education and training
- Incorporate national values, talent development and leadership values in curriculum development

Receive, consider, develop and review curriculum proposals

# CURRICULUM IMPLEMENTATION



## Introduction

In this lesson we define the term curriculum implementation and then discuss the strategies for curriculum implementation. We shall then outline the role of various agencies of curriculum implementation. We then examine teacher education in Kenya and finally discuss the challenges to curriculum implementation.



## Lesson Learning Outcomes

By the end of this lesson, you will be able to:

- discuss the strategies for curriculum implementation
- outline the role of various agencies of curriculum implementation
- examine teacher education in Kenya
- discuss the challenges to curriculum implementation



## : Definition of curriculum implementation and strategies to curriculum implementation

This is the national roll out of the curriculum in all applicable institutions. At this stage, the syllabuses and all necessary curriculum support materials are made available to the schools and other implementing institutions, and relevant guidelines on implementation are provided.

This is the stage at which the planned curriculum is introduced into the schools and colleges. It is the stage in which the newly developed and tried curriculum is made publicly available. This is the logical process to undertake after the tryout of the curriculum.

Implementation of new curriculum should only be attempted by the institutions (schools) in which the right conditions prevail. Implementation of a new curriculum should only take place in the institutions that have the right conditions. These institutions satisfactory arrangements have been made for:

- in-servicing of teachers
- purchase of adequate materials, teaching and learning aids and equipment
- necessary physical facilities

Implementation can hardly take place uniformly across the country or geographical area concerned. The schools should be grouped together according to their degree of readiness and implement the curriculum accordingly.

### **Strategies for Curriculum Implementation**

In summary form there are nine sub processes of implementation stages to be followed; (Oluoch, 1982).

#### **1. 1. Persuading people to accept the new curriculum.**

This begins at the conception and formulation of the curriculum development project. Many people in authority, teachers and parents should be persuaded to accept the new curriculum. Curriculum development and implementation is a team work and all team members must be persuaded to allow for smooth implementation

During implementation general information about the new curriculum started during piloting should be extended and intensified. Provision should be made for informing people and answering of questions and this should continue through implementation process.

Dissemination avenues include:

- Newsletters

- Magazines
- Newspapers
- Radio
- Television
- Social media (you tube, Facebook, Instagram, etc)
- School broadcasts
- Use of field officers (education officers, QASO, Curriculum support officers)
- Educating the teachers.

In-servicing of teachers started during piloting should be intensified and extended during implementation. For successful curriculum implementation the teachers involved must understand and accept the ideas contained in the new curriculum. Teachers have to understand, accept and internalize the philosophy behind the new ideas, materials, and teaching methodology in the new curriculum.

Teachers and teacher trainees in colleges and universities have to undergo through specially designed educational programmes

- Educating the teacher educators.

These include educational administrators, quality assurance and standard officers, college tutors, curriculum support officers and school heads.

Provision of necessary facilities and equipment.

Implementation of new curriculum requires suitable facilities and equipment such as classrooms, desks, seats, blackboards/whiteboards, playing fields etc. In addition, secondary schools will need laboratories, workshops etc. Some of these may require improvement and others may need procurement of new materials, for example textbooks or additional ones like exercise books etc.

Provision of these facilities and equipment is a joint responsibility of the educational authorities, school management and the community and parents.

- Supply of Curriculum Materials/and equipment.

Supply of curriculum materials becomes critical at this stage. Here we have the questions of

- Purchasing

This can be done by school management or by the government and then the materials are delivered to schools/collected from central points

- Delivery

Will depend on the procuring entity. School suppliers deliver to schools

- Storage

Books should be used with care and well stored when not in use to last at least four years.

- Production

This is done by:

- government owner firms

Kenya literature Bureau (K.L.B)

Jomo Kenyatta Foundation (JKF)

- Private companies. Oxford

University Press

East African Education Publishers

Moran Publishers

Longhorn

Publishers Focus

Publishers

## 1. Presentation of the new curriculum

This is the actual presentation of the new curriculum to the learners. Adequate preparations should be done to ensure proper presentation of the curriculum. This may include revision/reorganization of the timetable, in-servicing of teachers etc

Student assessment procedures/methods

These should be designed during curriculum planning Provision of continuous support for the teachers.

The head teachers and teachers will require continuous support from Quality Assurance and Standards Officers, Curriculum Support Officers, Educational Administrators and educational authorities.

## **Role of Various Agencies in Curriculum**

### **Implementation**

- TEACHERS SERVICE COMMISSION (TSC)

#### Functions

- Register trained teachers
- Recruit and employ registered teachers
- Assign teachers employed by the Commission for service in any public school or institution
- Promote and transfer teachers
- Exercise disciplinary control over teachers
- Terminate the employment of teachers
- Review the standards of education and training of persons entering the teaching service
- Review the demand for and supply of teachers
- Advise the national government on matters relating to the teaching profession



- **DIRECTORATE OF QUALITY ASSURANCE AND STANDARDS (MOE)**

**Functions.**

**It is responsible for**

- Establishing, maintaining and improving standards in all basic and training institutions;
- Quality Assurance and Standards assessment of basic educational and training institutions;
- Co-ordination, organization and implementation of co-curriculum activities at national and international levels;
- Vetting of expatriate teachers and institutions offering international curriculum;
- Liaison with KNEC on assessment of pre-service primary, ECDE and diploma teacher education examination teaching practice;
- policy formulation and implementation on matters related to Quality Assurance;
- Developing of assessment standards;
- Liaison with KNEC in moderation, awards and National Assessment and Learning Achievement (NASMLA) and examinations;
- liaison with Kenya Institute of Curriculum Development (KICD) on curriculum,
- design, development and evaluation of the national curriculum;
- Talent identification and development in schools and colleges;
- Advising the Minister and Permanent Secretary on all matters of Quality

Assurance and Standards in the country.



## : TEACHER EDUCATION IN KENYA

[Teacher Professional development: An International Review of literature](#)

Use the above link and read pages 43-54 on:

- Meaning of preservice teacher education
- Models of in-service teacher education
- The contents of preservice teacher education
- Challenges and limitations of teacher education

In-service Teacher education

[Teacher Professional development: An International Review of literature](#) Use the

above link and read pages 43-54 on:

- Meaning of in-service teacher education

## Types of in-service teacher education

- The contents of in-service teacher education
- Limitation of in-service teacher education

## What is In-Service Education?

Pre-service education takes place in residence in a college or University before a teacher is appointed to his first post or employed and registered by the Teachers Service Commission. On the other hand inservice education may be taken any time while the teacher is already trained and qualified who are also professionally employed. It may also be arranged for untrained teachers who have been recruited by T.S.C and registered to teach in public schools.

Whatever knowledge a teacher acquires during his initial training may not still be satisfactory in ten or fifteen years later. New knowledge keeps coming up through research and technology. In- service education is an essential element and condition for all teachers in public and private schools. **Role of in-service education**

- **Acquisition of New Knowledge**

Traditionally In-service education was popular with primary school teachers to prepare them to implement various changes in curriculum especially because they were not well academically prepared due to their short duration in training and the many subjects they took in college.

Graduate and diplomas teachers were regarded specialists. This view has changed with new development in education as witnessed through

programmes of one term or one year duration organized by overseas agencies for experienced serving teachers to attend advanced courses abroad for the teaching of Science, Mathematics and English etc

- Locally organized in-service programmes of one month or longer have been organized by the Ministry of Education in conjunction with the Kenya Institute of Education and the public universities which train teachers E.g. SMASSE
- In-service Education programmes for trained teachers of normal children have been organized in by KISE and the universities. These specialize as teachers for children with disabilities. Teachers of Art, Music, Drama and P.E. etc can obtain more knowledge and techniques through organized in-service education programmes.
- Teachers have done upgrading from P1 to diploma, Diploma to degree, Degree to Masters and masters to PhD in the recent years through:
- full-time programmes by enjoying study leave from the TSC,
- Continued Education Programme (CEP)/School based programme 🏫 Open and digital learning (ODEL)
- **Familiarization with Curriculum Development**
- To familiarize teachers with changes in curriculum development, seminars, workshops, and conferences are organized at sub-county, county and national levels.
- Changes in curriculum development are channelled through circulars and government documents from the MOE to the schools.

Heads of schools are expected to make such information available to teachers. School heads are expected to organize schools, seminars or meetings for members of staff to discuss new changes in curriculum.

- Training of examiners and markers for public examinations falls under this category. It is done by the Kenya National Examinations Council.
- **Familiarization with Principles of Organization and Management.**

The Ministry of Education, often organizes training courses ranging from diploma and certificate courses, seminars, workshops and meetings to familiarize head-teachers with essential principles of educational management. Such courses have become popular to teachers who got appointed to headship with a limited knowledge in educational management.

#### **Benefits of In-service Teacher Education to Teachers**

Teachers participate in in-service education for various reasons.

- **1. Promotion Status**

Many teachers in this country have been promoted from one grade to another after under-going an inservice education programme. The certificates awarded correspond to salary scales from other teachers in the profession. Promotions and changes of status are major reasons why most untrained primary school teachers attend in-service courses. The second reason would be to secure permanent and pensionable employment status enjoyed by other qualified teachers employed by T.S.C. after gaining knowledge in methods of teaching in the primary schools.

- **2. Improved Job Performance**

Changes in the teaching profession are common. Most teachers want to improve their performance to cope with new changes in education. To this group of teachers, promotions or salary are not reasons for participation. Curriculum Support Officers organize short in-service education courses for the improvement, increase and expansion of knowledge.

- **3. Increased Salary**

Salary in Kenya goes along with certificates and degrees obtained. Majority of in-service education programme may not be organized for the purposes of awarding extra or higher certificates or diplomas. However the following lead to increased salary

- Diploma or degree programmes for servicing teachers
- One year full time course for untrained graduate teachers. After successfully completing the course with salary increase or some adjustment (TSC does no longer employ untrained graduates).
- COMPETENCY BASED TEACHER EDUCATION
- A **Competency Based Teacher Education** programme specifies the **competencies** to be demonstrated by the student teachers. Specifically in relation to;
- Professional knowledge and understanding
- Professional skills
- Professional values and attitudes

- Attitudes

That facilitates and enhance the:

- Intellectual
  - Social
  - Emotional and
  - Physical growth of children.
- These competencies are inevitable for teaching effectiveness and therefore a student teacher must demonstrate them.

The essential elements of CBTE are:

- Teaching competencies to be demonstrated are role-derived and specified in behavioural terms.
- Assessment criteria are competency based – specify mastery levels.
- Assessment requires performance as prime evidence and takes student knowledge into account.
- The student teachers rate of progress depends on demonstrated competency.
- The instructional programme facilitates development and evaluation of specific competencies.
- The Competency Based Teacher Education approach focuses on what the student should know, be able to do and value as life-long learners. This therefore calls for development of competencies at multiple levels.

### **Generic Competencies**



- These competencies are anchored in the 21<sup>st</sup> Century skills and therefore represent the university's expectations of all graduates irrespective of the programmes they have pursued.
- These competences inform the expected learning outcomes of programmes.
- The competences also inform the student teachers learning experiences.

### **Programme Competencies**

- These competencies reflect distinctive characteristics of a graduate at the completion of the Teacher education programmes.
- The programme competencies are grounded on teacher professional standards and upon which the expected learning outcomes of the programme are framed.

### **Discipline/Subject Competencies**

- These competencies characterize the specific features of a discipline/subject of a programme.
- They articulate the specific competencies in terms of knowledge, skills and attitudes in

relation to the

discipline. **Course/Unit**

### **Competencies**

- These competencies inform the outcomes that the student teachers will demonstrate achievement by the end of the course according to predetermined standards of performance and content.
- Outcomes based on course competences should be explicit and should guide the teaching learning and assessment activities of the course/unit.

- The competencies exist in an iterative relationship with the generic, programme and discipline competencies



## Challenges to Curriculum Implementation

Challenges affecting effective implementation of basic education include:

- insufficient physical facilities and instructional resources to cope with the exponential growth of student population resulting introduction of Free Primary Education and Free Day Secondary Education;
- inadequate teachers resulting in high teaching load prompting the use of ineffective teaching methods;
- Lack of motivation of the teaching force resulting into insufficient focus on the learner and thus creating little room for use of modern teaching techniques that require individualized teaching, amongst others.

## CURRICULUM EVALUATION



### Introduction

In this lesson we define the terms curriculum evaluation, assessment of learning, assessment as learning, assessment for learning, criterion-referenced assessment, norm-referenced assessment and assessment rubric. We then discuss the role of Formative evaluation (assessment for learning, assessment as learning) and summative evaluation (assessment of learning). We then explain the

principles of instructional evaluation. Then the evaluation activities at each stage of [curriculum development process](#) are examined.

## Lesson Learning Outcomes

By the end of this lesson, you will be able to:

- define the terms curriculum evaluation, assessment of learning, assessment as learning, assessment for learning, criterion-referenced assessment, norm-referenced assessment and assessment rubric
- discuss the role of Formative evaluation (assessment for learning, assessment as learning) and summative evaluation (assessment of learning)
- Explain the principles of instructional evaluation

Examine the evaluation activities at each stage of [curriculum development process](#)

## Definition of Key concepts in curriculum evaluation

**Curriculum** evaluation refers to the process of studying the merit or worth of some aspect or whole of a curriculum. It is concerned with —what| and —how| aspects.

Curriculum evaluation is the process for assessing the effectiveness of changes that have been made. It demonstrates the effectiveness of a curriculum. Component or program

**Evaluation** in the content of education is a process used to obtain information from testing, from direct observation of behaviour, from essays and from other devices to assess a student's overall progress towards some predetermined goals or subjects. It includes both a qualitative and quantitative description and involves a value judgment of overall student behavior for decision making.

Evaluation and measurement are not the same, although evaluation involves measurement. If we assess a student's knowledge and understanding in a subject by means of an objective or essay type test, that is measurement. If a teacher puts a value on the student's work, talents, attitudes and other characteristics of behaviour that is evaluation. Evaluation should in part involve testing that is nonsubjective on the part of the teacher, otherwise it is likely to be unreliable.

Assessment is used interchangeably with evaluation. Testing is the process of using an instrument or test to measure achievement. Measurement and testing are thus ways of gathering evaluation and assessment data.

### **Monitoring and Evaluation**

Monitoring and evaluation are curriculum management functions through which the Institute establishes the progress of [curriculum implementation](#) programmes and the extent to which curriculum objectives are achieved.

Monitoring is a continuous undertaking that informs the Institute of any bottlenecks in the implementation process for purposes of carrying out corrective interventions in order to ensure the curriculum objectives are met. Necessary interventions are made during the implementation process based on the monitoring results. Regular monitoring is undertaken throughout the lifetime of [curriculum implementation](#).

Evaluation is a time-bound activity which is carried out to determine the relevance, performance and success of curriculum programmes either during the implementation process or at the end of a definite cycle. Evaluation establishes the extent to which the curriculum objectives have been achieved and the overall success or failure of the curriculum programmes in terms of expected

outcomes. Results of monitoring and evaluation may lead to curriculum change, reform or innovation.

The curriculum is then maintained after all these series. In maintaining the programme, the curriculum leaders strive to stabilize it and keep operational the content; experiences and environments. The maintenance requires a steady flow of accurate data or information in order to assess continual programme performance. It means managing the curriculum and support systems.

### **COMPETENCY BASED ASSESSMENT**

competency based assessment is a key reformation which will facilitate the adoption of formative assessment practices that promote diagnostic approaches which will in turn enhance learning and improve learning outcomes. The new curriculum, which is competency based emphasizes on utilization of formative assessment as the basis for improvement of teaching and learning. The continuous assessment tests set to replace one off examinations will be standardized and administered at various levels.

This is a departure from current assessment practices that seek to compare learners with each other (assessment with reference to a normal distribution, or norm referenced) and is a shift towards assessment practices that seek to collect evidence and make judgments on the extent and nature of progress towards a learner's achievement.

- Validity
- Reliability
- Fairness

Flexibility

- Access

### **Formative Assessment (Assessment for Learning)**

The goal of formative assessment is to monitor student learning to provide ongoing feedback that can be used by teachers to improve pedagogical strategies and by students for improvement in their learning.

The assessments should be made on recorded evidence based on anecdotal records to be maintained by the class teacher or subject teacher.

The overall assessment should be followed by descriptive remarks by the class teacher or the teacher responsible for the subject about positive and significant achievements, avoiding negative assessment even by implication. Formative assessment can be carried out using multiple modes of assessment

### **Summative Assessment (Assessment of Learning)**

Summative assessments are used to evaluate student learning, skill acquisition, and academic achievement at the conclusion of a defined instructional period – typically at the end of a learning cycle – by comparing it against some standard or benchmark. In Kenya, summative assessment will be at the end of: upper primary education, lower secondary education and senior school. Summative assessment is used for placement and certification

### **Assessment Instruments for Formative Assessment**

- Observation

## Observation schedule

- Checklists,
- Rating Scales and
- Rubrics
- Questionnaires
- Project Method
- Journaling
- Portfolio
- Question and

## Answer **Competencies**

- Knowledge and understanding: Does the child demonstrate an understanding of the subject?

Has the child mastered the key subject concepts?

- Practical skills: How does the child perform on aptitude and practical situations?
- Attitude and values: How does the child respond to a task or a situation? What is the child's behaviour like in a range of situations and contexts?
- Generic competencies: What are the steps taken to perform a given task? What is the reasoning behind them? How does the child overcome each challenge?
- Knowledge and understanding will have indicators such as correctness of answers, coherence of ideas, and logical reasoning.

- Practical skills will have indicators such as accuracy, using appropriate methods, quality product, speed and efficiency, and coherence.

Attitude and values will have indicators such as approach to a situation, appreciation of the task given, impression of a situation, manipulation, reasoning, persistence, and tolerance.

### **Performance Indicators**

- Knowledge and understanding will have indicators such as correctness of answers, coherence of ideas, and logical reasoning.
- Practical skills will have indicators such as accuracy, using appropriate methods, quality product, speed and efficiency, and coherence.
- Attitude and values will have indicators such as approach to a situation, appreciation of the task given, impression of a situation, manipulation, reasoning, persistence, and tolerance.
- Generic competencies will have indicators such as reasoning, manipulating, presenting, value judgment, and applying knowledge.

### **Grading of competency based assessment/Rubric**

Excellent :This is a standard of excellence level. Descriptions should indicate that all aspects of competencies exceed grade level expectations and show exemplary performance or in-depth understanding. Learning goals are met in a comprehensive way.

Very Good :This is approaching the standard of excellence level. Descriptions should indicate some aspects of work that exceed grade level expectations and demonstrate solid performance or understanding. Learning goals are met in a practical and thorough way.



Good: This meets acceptable standards. This level should indicate minimal competencies acceptable to meet grade level expectations. Learning goals are met in an appropriate and reasonable way.

Sufficient: Performance and understanding are emerging or developing but there are some errors and mastery is not thorough.

Not sufficient: This does not yet meet acceptable standards. This level indicates what is not adequate for grade level expectations and indicates that the student has serious errors, omissions or misconceptions. The teacher needs to make decisions about appropriate interventions to help the student improve.

## **Role of Formative Evaluation (Assessment for Learning, Assessment as Learning) and Summative Evaluation (Assessment of Learning)**

### **PHASES OF EVALUATION AND PURPOSES OF CURRICULUM EVALUATION**

There are three phases of evaluation which every teacher needs to know.

- Pre-assessment
- Formative Evaluation
- Summative Evaluation

These terms are technical words to differentiate evaluation that takes place before instruction (preassessment), during instruction (formative), and after instruction (summative).

Pre-assessment evaluation is provided before instruction that takes place before instruction to determine the starting point on an instructional program. It identifies need prerequisite skills and causes of learning difficulties and to place students in learning groups. Formative evaluation consists

of those techniques of a formal and informal nature, including testing, that are used during the period of instruction. Progress tests are given in the classroom are a good illustration of formative evaluation.

Thorough formative evaluation teachers may diagnose student difficulties and take remedial action to help them overcome their difficulties before they are confronted with the terminal (summative) evaluation. Formative evaluation enables teachers to monitor their instruction so that they may keep it on course. It is also used to provide assessment of curriculum quality. It is conducted during the [curriculum development process](#) for the additional purpose of providing information that can be used to form a better finished product. Thus formative evaluation takes place at a number of intermediate points during [curriculum development process](#).

Summative evaluation is the assessment that takes place at the end of a course or unit. A final examination (post-test) means used for the summative evaluation of instruction. Its major purpose is to find out whether the students have mastered the preceding instruction.

A good teacher utilizes results of summative evaluation to revise his or her program and methodology for subsequent groups.

## **PURPOSES OF INSTRUCTIONAL EVALUATION**

- **Evaluation as a basis for school marks or grades by teachers**

Educational systems require that teachers occasionally submit marks or grades on students. These marks or grades can be arrived at through formal examinations, regular tests, assignments, laboratories reports, observational information or combination of these.

- **Evaluation as a Means of Informing Parents of schools children** have a right to know how their children are progressing in school. Students' evaluation is the most important way of providing them with this information.

- **Evaluation for Promotion to Higher Class**

Student evaluation is sometimes used to determine whether a student has made enough progress to be promoted to a higher class or form in the school.

- **Evaluation for Student Motivation**

Success in tests and examinations as well as sports and other school activities can give great encouragement to students. Similarly, failure to do well can make students work harder or strive to do better. Teachers should try to give their students feedback on the

evaluation of all aspects of their learning and behaviour so that both those who do well and those who do not will be motivated to improve on their performance.

- **Evaluation for Guidance and Counseling Purposes**

All students need to be advised to help them solve their own personal problems, whether academic or emotional. The two types of problems are indeed often connected.

Successful students tend to enjoy school more than those who are not so successful. It is generally those students who appear to be failure as indicated by the evaluation of the teaching staff, who need the most attention and it is to them that the class teacher must direct herself.

- **Evaluation to Assess the Effectiveness of the Teaching Strategy**

If a teacher does not in some way assess the students' improved knowledge, understanding and higher cognitive skills as well as their attitudes and psychomotor abilities, will not be able to evaluate the success or otherwise of the teaching strategy she has employed. A higher failure

rate in a course is more often due to poor teaching than to the lack of intelligence of the students.

- **Evaluation for Employment Purposes**

Not all students who pass through post-primary schools will proceed to university or other institutions of higher learning. Some students may decide to join a company or business. Employers normally require information on potential employees with reference to academic ability, attitude to work, moral character personality and so on. It is therefore, necessary for the teachers to evaluate nearly all aspects of the students' performance while they attend the school.

## **2. Curriculum Evaluation**

The primary purpose of curriculum evaluation is, of course, to determine whether the curriculum goals, and objectives are being carried out. These goals and objectives are to be evaluated in the first place to determine if they are the right kind of objective. It also determines whether the curriculum is functioning while in operation, and using the best materials and the best methods.

Curriculum evaluation also determines whether the products of our schools are successful in higher education and in jobs, whether they can function in daily life and contribute to our society. Curriculum evaluation also determines whether educational program is cost- effective, that is to say whether the people are getting the most of their money.

### **Role of Evaluation in Curriculum**

📖 Evaluation entails not a single study but also a series of sub-studies, performed at the various stages of [curriculum development process](#). The main aim of curriculum evaluation is to collect descriptive information about an educational programme, which is then used to modify and improve the program; to compare the program to other programs, and to make judgement as to the worth of the program or project.

Evaluation generally answers the questions regarding: selection, adoption, support, change, innovation and worth of an education program. Formative evaluation helps the curriculum at the planning stage to formulate a new educational policy, to discover new needs, to formulate new goals, to clarify in their definition of goals, select appropriate goals from competing goals, determine complementary goals, number of new disciplines to be taught, provide empirical evidence on the view of teachers, parents, learners, employers, professional on what requires change or towards the envisaged change.

At the project materials development stage, evaluation helps to translate broad educational goals into curriculum objectives for various subjects, re-define general objectives into operational objectives, determine appropriate syllabus and textbooks structure and content; ensure relevance in content scope, sequence, continuity, integration and balance, identify areas that require in-service for teachers and determine appropriate evaluation procedures and process.

At the project field-trial (try-out) stage evaluation helps to determine:

- ♦ Adequacy of the materials
- Needed revisions, cost of production, quantity
- Nature needed
- Teaching-learning strategies need
- Evaluation procedures – instrument required

Evaluation at the school implementation stage checks on a range of teaching-learning

- resources in schools and their use by teachers and students
- Whether teachers have been adequately prepared
- Reaction of teachers, parents and other stakeholders on the new program
- Whether procedures used meet the standard criteria designated by the curriculum developer
- What is actually being learned – the operational curriculum
- What requires change
- Acceptance of the new program by the beneficiaries

National examinations cater for comparison of schools, districts, pupils, teachers. Grading selection, placement, certification, school academic records. Evaluation of

- examination procedures ensures higher standards of education, detects areas of difficulty; determiners new methods of teaching and evaluation; problems teachers and learners have in interpreting course objects.



## Principles of Instructional Evaluation

- **PRINCIPLES OF INSTRUCTIONAL EVALUATION**

Instructional Evaluation should be used on the following criteria by which worth is determined.

- **Consistency with Objectives**

Evaluation should be used to measure what is indicated in curriculum objectives of a course programme R. Tyler (1950) observer that educational objectives are the criteria by which materials are selected, content is outlined, instructional procedure are developed and tests and examinations are prepared. Evaluation tells us how successful we have been in this effort. The criteria for evaluation and the results so obtained should be underscored and accepted by all those concerned. There is need to develop in pupils the ability to learn further.

- **Validity and Reliability**

Evaluation instruments are valid if they measure what they are supposed to measure. A test in CRE, for example, should not be expected to elicit scientific knowledge but religious education concepts. A valid test shall relate to objectives of the specific course and appropriate for the level.





- Reliability refers to the consistency with which an evaluation instrument measures giving the same score of results. Two different examiners are able to arrive the same score on the test candidates who have gone through the same learning process. A test item that has several answers yet the test constructor required only one correct answer cannot be said to be reliable.

A reliable test will also try to elicit the same abilities, skills from the same sample of students.



- **Continuity**

Curriculum evaluation should be an on-going process in order to provide effective feedback, which will lead to course improvement. It should moreover relate to previous, present and future learning experiences and follow proper sequencing of the course, from easy to complex items. The evaluation system begins with curriculum decision, which results in the identification of the first goals. It continues throughout the planning process into implementation activities, and cycles back to the planning process.

Instructional evaluation should be a continuous process so that the teacher can adequately and effectively assess each student needs in order to select appropriate resources, develop appropriate learning strategies, judge each student merit, and provide effective feedback and motivation to each student; plan group methods, appropriate activities along specified learning objectives.

Through continuous assessment the teacher consistently and systematically provides the educational experience most suited to the educational needs, interests, readiness and ability of each student.

- **Balance**

Balance means that the curriculum developers have weighed the relative importance they have given to each student need and development tasks. Considerations should be given to all student needs.

Evaluation should assess all skills weighted against the time allocated to each. Balance also ensure that the various cognitive skills are equally or reasonably weighted. If the test items dwell on higher or lower cognitive skills only, such a test will lack balance. Hence, there should be a balance in the following cognitive levels of skills: knowledge, comprehensive, application, analysis, synthesis and evaluation. There should be also balance theory and practical skills. All subjects have theoretical and

practical aspects.

- **Comprehensiveness**

Education aims at the development of the whole person. Hence, all the objectives of the curriculum programme should be evaluated, namely: the cognitive, effective, psychomotor, spiritual and social relating domains. Evaluation instruments should be designed to yield accurate information concerning personal, social adjustment, physical growth, spiritual growth, habits of work, interests and attitudes; special aptitudes, growth in creative ability, home and community backgrounds must also be available if the school is to do the best job possible in fostering the wholesome growth of learners and preparing them for effective citizenship in a democratic multi-part, multi-racial, multi-ethnic, multi-religious society such as obtains in Kenya. Education should prepare the individual to face the vicissitudes of life with constancy, persistency, insistence and courage.

Individuals play different roles at different educational experience and stages of life; Pre- primary, primary, secondary, university; childhood, adolescence, adult middle age, retirement age and old age. Each stage requires definite knowledge, skills, values and are subject to evaluation by society. It is not the mere acquisition of knowledge that matters

but how it utilized. Modern evaluation attempt to obtain as complete a picture as possible of the individual. The evaluation procedure is comprehensive if they utilize a variety of means and techniques in collection evaluation data.

- **Cooperation**

An evaluation system is depended upon the adequacy of the planning which resulted in the selected or creation of curriculum programs, activities, procedures, resources and other elements to be evaluated. Evaluation systems are also dependent upon the utility and integrity of the specific data to be gathered, displayed and interpreted (practical skills, work at primary, secondary, and university exams). What criteria should be met to ensure validity and reliability?

An adequate evaluations system involves at least two levels of cooperation. The first concerns the integrity of the relationships established among the planning, implementing and evaluation phases of program building. Evaluation should be inbuilt in the planning and preparation states of curriculum development. The second concerns the comprehensive involvement of all parties, which have legitimate input or from the programs activities or its evaluation systems.

The determination of what constitutes success or failure requires the cooperative involvement of those who implement and are affected by the program and those who evaluate. There should be cooperation among the KIE curriculum developers, the Kenya National Examinations Council and the classroom teachers. There should be also cooperation among the psychologists, sociologists, philosopher, religious leaders, professors, trade unionists, curriculum developers, employers and teachers; and all other stakeholders.

Evaluation instruments should be functional, practical understood and acceptable by all teachers involved. There should be a closer relationship between the examination and the objectives which

school education hopes to achieve, therefore, between the style of the What is the difference evaluation and measurement?

# CURRICULUM INNOVATIONS



## Introduction



### Introduction

In this lesson we define the term curriculum innovation, and then discuss innovation models. The criteria for curriculum innovation will then be explained. Key recent Curriculum Innovations in Kenya will be examined and finally challenges to curriculum innovation will then be discussed.



## Lesson Learning Outcomes

By the end of this lesson, you will be able to:

- Explain Why curriculum innovation/change?
- Discuss innovation models
- Explain the criteria for curriculum innovation
- Examine key recent Curriculum Innovations in Kenya

Discuss challenges to Curriculum Innovation



## Why Curriculum Innovations/Change

Meaning of curriculum innovation and change

- **Innovation** is defined as activating forces within the system to alter it.
- **Innovation or curriculum change** is an effect of change where educational institutions adjust to changes occurring to other social institutions. The curriculum has to adjust to the economic, technological and ideological factors
- Innovation or curriculum change can also be described as a condition of change where changes in curriculum or education for that matter are necessary for broader social changes.
- Curriculum change must be planned.
- **Is curriculum change the same as curriculum improvement?**
- **Curriculum improvement (Taba, 1962)** is concerned with discrete aspects of the curriculum without fundamental changes in its conceptual design or organization. On the other hand curriculum change or innovation in a way means to change an institution affecting several aspects.
- **Change can be in three levels**
  - Minor changes may involve the re-arrangement of the sequence of the subject matter or learning activities or merely the addition of a topic or method to instructional programme.
  - Medium changes may involve an innovation like the integration of subjects, a new subject or a new approach to the existing subject
  - Major changes will affect many aspects of the curriculum-content, methodology, approaches, materials, subtracting and adding to what is existing. This may also involve changes in the conceptual design and the organization thus calling for a new plan.

It is important to note that change usually aims to improve the curriculum. However, change may not necessarily bring improvement but the objective is to improve the existing programme in the context of the new environment.

## **.2.2 Models of Curriculum Innovations**

Ronald Havelock (1969) of University of Michigan identified three main models of innovation.

- The Research , Development and Diffusion Model (R,D &D)
- The social interaction model
- The problem-solving model
- **TheResearch,DevelopmentandDifusionModel(R,D&D)**

The idea or practice is conceived at a central planning unit (head) and then fed into the system.

It is also called centre-periphery approach.

It involves the following

- Basic research by a central project team which develops a new curriculum, devices and designs prototype materials etc
- Field trials- the development and testing of prototype materials in a number of

trial schools, followed by re-design and revision in the light of these trials

Mass production of the modified prototypes

- The planned mass dissemination of the innovation by conducting courses, conferences, workshops etc
- Implementation of the innovation by the users (the schools, teachers, pupils).

#### **Advantages.**

- Can harness the efforts of experts and talented teachers to produce materials of high quality
- Such a system can ensure that national priorities are given adequate emphasis
- Curriculum development centres can take the leading role on principles of curriculum development in the crucially important task of training and re-training of teachers for their new tasks
- Materials are tested before diffusion on a large scale.

#### **Disadvantages.**

- High initial development costs though compensated by resulting quality and efficiency in the long term

- Teachers are involved in the development process only to a limited extent, otherwise they are passive recipients of the curriculum
- Because of the high degree of centralization there is a danger of failing to take into account local needs and variations

Not suitable for new strategies for teaching though very effective for production of materials

- **b. The Social Interaction Model (SI)**

- According to this model, change proceeds through contacts, formal and informal, among interested individuals and groups of people.
- Social interaction is the usual way by which ideas and practices are diffused through society

-by informal contacts between interested individuals and groups.

- The SI model emphasizes diffusion, the movement of messages from one person to another and system to system
- It stresses the importance of interpersonal networks of information, of opinion of leadership, personal contact, and social integration.



- The strategy takes the form of convincing a respected administrator or teacher of the usefulness of a new device or practice, and then enabling colleagues to come and see for themselves the new practitioner using the innovation.

### **Advantages**

- A natural process adopts a periphery-periphery approach  
Though it is an unplanned approach, it can be made more systematic by structuring and coordinating the contacts between groups and individuals interested in curriculum

development. This can be done through-courses, conferences, visits to schools, by QASOs etc. In this case, the central Agency acts as a co-coordinator or communicator of ideas

### **Disadvantages**

- It involves individuals rather than groups or organizations while it is the latter that have to implement the innovations
- It is informal and unplanned and thus can be slow.
- **c. Problem Solving Model (PS)**
- Individuals are themselves involved in conceiving, initiating and developing innovations at local level. It is a periphery-centre-periphery approach to innovation.
- It is based on the assumption that innovations is part of a problem solving process which takes place inside the user or client system, be it a school, the teacher or the student.
- Here the initiative comes from the periphery (the school or client). Innovators at the periphery set about solving their own problems. The model is built around the user of the innovation.
- It assumes the user has a definite need and that innovation satisfies that need.
- Thus the process is from problem to diagnosis of a need then to trial **Steps**
- A need is identified

- The need is translated into a problem which is then diagnosed
- The diagnosis leads to a search for solutions
- The possible solutions are evaluated. The innovation which seems to provide the best

solution is then tested for its effectiveness.

### **Advantages.**

- The innovations are initiated, generated and applied by the teachers and schools themselves on the basis of their needs. Thus they have a stronger user commitment and long term survival
- It is flexible enough to encompass all types of innovations-materials, methods, groupings of pupils etc,
- The chosen innovation (solution) is geared to the particular circumstances of the school/client

### **Disadvantages**

- Being local and limited in size, the quality of the innovation e.g., materials may not be as good as that of R,D&D
- Though appropriate to the needs of the school, the particular innovation may not be based on sound theory or practice
- Such innovation usually take up a considerable amount of the teachers time.

- **d. The Linkage Model**

Due to the weaknesses of the three models, Havelock has proposed a fourth model which combines aspects of all the three models by using linkage procedures and agencies intermediate between the centres of the curriculum and schools, which mediate and link all the parties involved in the innovation process

## **7.2.3 Criteria for judging the quality of an Curriculum**

### **Innovation**

Criteria for evaluating curriculum innovation

Rogers and Shoe maker (1971) identifies the following criteria for judging a curriculum evaluation

- Relative advantage

This is the degree to which a curriculum is perceived as being better or introducing something more useful will determine its acceptability and rate of adoption

- Improve the students learning
- Is economically manageable
- Low initial cost
- Posses no special risk to anybody's job security

- Compatibility

Refer to the degree to which an innovation is perceived as being in agreement with or supportive of the potential adopters needs and values. How will the new idea or practice fit into or support the existing educational programme.

- Triability

This relates to the extent to which an innovation can be piloted, without taking too much time, energy or funds. Can a limited trial produce sufficient data to make a decision whether the innovation is necessary?

- Observability

The potential adopters and the sponsors would like to see tangible results from their efforts.

- Immediate consequences
- Accountability-availability of concrete data

- Complexity

**An innovation must be simple enough to be understood and utilized. If the potential**

## **adopters perc 7.2.4 Challenges to Curriculum**

### **innovations**

Challenges to curriculum innovations

- Social and demographic factors Most people will resist change
- If it threatens their cherished traditions and institutions
- Due to personal reasons as for example where such change affects them or their positions negatively
- Because they do not approve of the source of the new idea or they may anticipate adverse effects on the society

Public institutions, organizations or private agencies can resist change, if they deem it detrimental to their well-being or to the well-being of society as a whole.

- Economic factors

Most curriculum innovations come about due to certain economic problems.

- A curriculum that may be seen to contradict the official economic policies is unlikely to succeed.
- On the other hand, curriculum changes either succeed or fail depending on the availability of funds for:
  - training of teachers,
  - inservicing teachers,
  - materials production,
  - textbooks,
  - more facilities in schools,
  - workshops seminars,
  - experiments,
  - research
- Politics
- A good relationship must exist between the curriculum specialist and the contemporary government, because their government can either promote or block any curriculum project which is not in line with her policies and which does not fit in the framework of the political ideology.
- Frequent changes in government make it more difficult to maintain a consistent policy

and to follow up plans.

- Technological factors
- Schools are always trying to change their programmes to cope with new technology
- Many curriculum projects have been overtaken by events due to rapid technological changes.
- Schools are also becoming a big market for new inventions
- Use of electronic machines such as computers and calculators in various aspects of life , including education, puts a lot of pressure on the curriculum
- Epistemological factors

The constant change and high rate of expansion in knowledge puts pressure on the curriculum specialist-what content to include and what to leave out. Considerations have to be done bearing in mind the changing nature of knowledge, learners, more knowledgeable teachers.

- Physical factors

In developing a curriculum for school the prevailing or possible future physical conditions in the schools/country must be put into consideration.